SIMULATING THE DAYTIME SENSITIVITY OF INTERVERTEBRAL DISCS

Laura Baumgartner[1], Jérôme Noailly[1]

[1] BCN MedTech, Pompeu Fabra University, Spain, laura.baumgartner@upf.edu

Introduction

Intervertebral disc (IVD) degeneration is a major risk factor for low back pain. The IVD is exposed to high loads as it articulates the spine. Due to organ avascularity, IVD cells live within a hostile nutritional environment. While the impact of mechanical loads and nutrition have been widely studied, daily oscillations of tissue sensitivity to stimulation (daytime sensitivity, DS) caused by circadian rhythms, are largely unknown, despite its recognized importance [1]. This work introduces a mathematical framework that allows to approximate the effect of DS directly on the targeted mRNA expressions (exp). being the main structural proteins Aggrecan (Agg), Collagen Types I, II (Col-I, Col-II) of the IVD's central tissue, the Nucleus Pulposus (NP).

Methods

A previously developed parallel networks (PN) methodology for NP cell activity predictions [2] was extended to accept DS. The PN-Methodology interprets mRNA exp as interconnected small, directed parallel networks. PN are fed with in vitro measurements of cell responses to varying exposure to stimuli. Each link is determined by two normalised variables, θ and x reflecting the importance (θ), and the dose (x) of a stimulus for an mRNA exp (Fig.1). θ is scalar that varies between 0.001 (low importance) – 1 (high importance). To simulate DS, θ of Agg and Col-I/Col-II was set to 0.3101 and 1, respectively, according to experimental knowledge [3]. x was informed by observing experimental dynamics of DS tested by applying a compressive load at four different moments along the day [1] (Fig.2). Dynamics were approximated by a combination of basic mathematical concepts generally used to explain natural phenomena; circadian oscillations were represented as adapted sine functions with a period of 24h. Phase shift and amplitude were modulated using the sine-derived slope and decay functions. A decline of amplitude in absence of stimulation was directly derived from [1]. Col-I was assumed to oscillate in antiphase to Agg and Col-II, because of its opposite function to Col-II in the NP. The integrated model was verified against its capacity to reproduce phase shifts and amplitude changes found by [1].

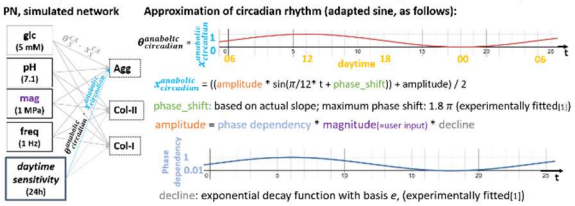


Figure 1: Simulated network (input conditions in brackets) with illustration of (anabolic) daytime sensitivity ($x^{anabolic}_{circadian}$) and phase dependency. mag: magnitude, freq: frequency, glc: glucose, CA: cell activity. S: Stimulus

Results

Simulations correctly predicted a phase shift (advance/ delay) for stimulation at mid descending/mid ascending phase. Stimulation at peak resulted in highest amplitude, and no phase shift, stimulation at trough disrupted the CR of Agg and Col-II (Fig.2). mRNA downregulation at the moment of impact was reproduced.

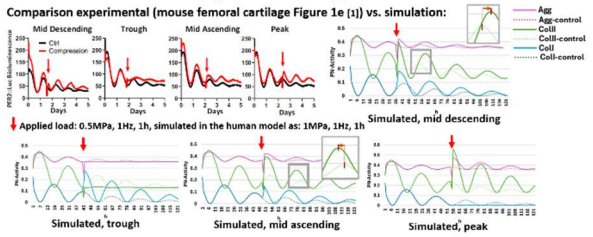


Figure 2: Experimental (top left) [1] vs. simulations. Phase shift visualized in zoom.

Conclusions

The model largely reproduced DS observed in experimental research (Fig.2). This indicates that basic mathematical formulations might be a useful tool to approximate highly dynamic cell responses. Thereby, a direct link between the cellular (micro-) environmental conditions and eventual cell responses was made. This reflects a time efficient approach, allowing to simulate long time periods within seconds. The model predicts overall highest cell activity for Agg, followed by Col-II and Col-I, which reflects the relative NP tissue composition. Predictions for mRNA downregulation at impact could only be reproduced using 1 MPa of impact load in the human model, vs. 0.5 MPa in [1], which suggests that 0.5MPa reflects a high load for mouse cartilage. Per2 and Agg/Col-II are activated through the SMAD2/3 pathway, based on which similar Per2-Agg/Col-II dynamics were assumed [4]; and high similarity between cartilage and IVD cell responses [1] proposes cartilage to be a good model to approximate IVD dynamics. This is the first model that allows to investigate highly dynamic daytime sensitive IVD regulations. Future work shall extend the model to simulate multiple impacts to approximate human activities over months. This would allow (e.g.) to shed light on gene exp dynamics of shift workers.

Acknowledgements

ERC-2021-CoG-O-Health-101044828.

References

1. Dudek M et al, Nat Commun 14,7237, 2023.
2. Baumgartner L et al, Heliyon 12, 2026.
3. Dudek et al, J Clin Invest. 126, 2016.
4. Workineh et al, Phys Life Rev 57, 2026